8.7 Buffer Solution

Do you know that our blood maintains a constant pH, irrespective of a number of cellular acid – base reactions. Is it possible to maintain a constant hydronium ion concentration in such reactions? Yes, it is possible due to buffer action.

Buffer is a solution which consists of a mixture of a weak acid and its conjugate base (or) a weak base and its conjugate acid. This buffer solution resists drastic changes in its pH upon addition of a small quantities of acids (or) bases, and this ability is called buffer action. The buffer containing carbonic acid (H_2CO_3) and its conjugate base HCO_3^- is present in our blood. There are two types of buffer solutions.

1. Acidic buffer solution : a solution containing a weak acid and its salt.

Example : solution containing acetic acid and sodium acetate

2. Basic buffer solution : a solution containing a weak base and its salt.

Example : Solution containing NH_4OH and NH_4Cl

8.7.1 Buffer action

To resist changes in its pH on the addition of an acid (or) a base, the buffer solution should contain both acidic as well as basic components so as to neutralize the effect of added acid (or) base and at the same time, these components should not consume each other.

Let us explain the buffer action in a solution containing CH_3COOH and CH_3COONa. The dissociation of the buffer components occurs as below.

$$CH_3COOH\,(aq) \rightleftharpoons CH_3 - COO^-_{(aq)} + H_3O^+(aq)$$

$$CH_3\,COONa\,(s) \xrightarrow{H_2O_{(l)}} CH_3 - COO^-_{(aq)} + Na^+(aq)$$

If an acid is added to this mixture, it will be consumed by the conjugate base CH_3COO^- to form the undissociated weak acid i.e, the increase in the concentration of H^+ does not reduce the pH significantly.

$$CH_3COO^-(aq) + H^+(aq) \longrightarrow CH_3COOH\,(aq)$$

If a base is added, it will be neutralized by H_3O^+, and the acetic acid is dissociated to maintain the equlibrium. Hence the pH is not significantly altered.

$$OH^-(aq) + H_3O^+(aq) \longrightarrow H_2O\,(l)$$

$$CH_3COOH\,(aq) \overset{H_2O\,(l)}{\rightleftharpoons} CH_3COO^-(aq) + H_3O^+(aq)$$

$$OH^-(aq) + CH_3COOH\,(aq) \longrightarrow CH_3COO^-(aq) + H_2O\,(l)$$

These neutralization reactions are identical to those reactions that we have already discussed in common ion effect.

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